

Algorithmic Methods

December 2024

Exercise I - Data Types

Consider the following Python statements:

```
1 exampleText = "This is an example"
2 grades = [5,4,3,1,6]
```

1. To which data type does the *exampleText* variable belong?
2. Which data type is returned when the *split()* method is applied to *exampleText*?
3. By which index can the first element of *grades* be accessed?
4. By which index can the last element of *grades* be accessed?
5. By which slice can the last two elements of *grades* be accessed?

Exercise II - Loops and Conditions

a)

Consider the Python program below:

```
1 highest = 0
2 grades = [5,4,3,1,6]
3 for grade in grades:
4     if grade < highest:
5         highest = grade
```

After executing this program, what is the value of *highest* ?

b)

Consider the Python program below:

```
1 exampleText = "This is an example"
2 lowered = []
3 for word in exampleText.split():
4     lowered.append(word.lower())
```

Write down the contents of *lowered* after two iterations!

Exercise III - Bughunter

Consider the program below:

```
1 def posSubtract(x,y)
2     if x<Y:
3         return y - x
4     else:
5         return x-y
```

There are three errors in the code above. Mark all errors and explain how to fix them!

Exercise IV - Code Execution

Consider the programs a) and b) below:

a)

```
1 words = ["finding","ending","end"]
2 for word in words:
3     if word.endswith("ing"):
4         print (word)
5     if word.startswith("end"):
6         print (word)
```

b)

```
1 words = ["finding","ending","end"]
2 for word in words:
3     if word.endswith("ing"):
4         print (word)
5     elif word.startswith("end"):
6         print (word)
```

Write down all outputs for both programs!

Exercise V - Regular Expressions

Have a look at the program below and write down all outputs!

```
1 import re
2 numbers = [1,5,7.5,115.555,.66]
3 for number in numbers:
4     if re.search(r"^\d*\.\d+$",str(number)):
5         print (number)
```
